

Objective:

To construct a predictive model and analysis to choose whether the customer will buy or not.

Use the Python codes and include the diagrams in the report. Make sure to edit the colours to make your work unique. Also interpret the results from the codes to help in the decision-making process.

See here for the codes:

<https://medium.com/analytics-vidhya/predictive-web-analytics-a-case-study-f30feda45002>

See here for the data:

<https://archive.ics.uci.edu/dataset/468/online+shoppers+purchasing+intention+dataset>

CASE STUDY

A e-Company that has employed smooth, connected methods using a new platform with the objective of acquiring improved and advanced comprehension into operational performance and constructing enhancements for better results. The organisation has gone through problems curtailing from frequent breakdown and failure, including long and unintended interruption and expensive unexpected expenses– which leads to losses and unhappy clients.

Its website is linked and company desires to detect plans and patterns that indicate appropriate operation and other operational performance pointers. These understandings will help to enhance planning, decision-making, and operations results.

The website produces data. The organisation is trying to search for analytical solutions and strategies that will observe, analyse, manage, and process this information to answer the following in your report.

- Recognise and identify performance patterns and trends that will help the organisation and clients.
- Give effective decision making strategies for the company.
- Forecast and predict unsolicited events for e.g. failure or required methods

- Build a predictive model using client visits data on a website.

Dataset Information:

The dataset contains data points of 12,330 customer session visits to the website. The dataset was designed so that every single session would fit in a different user in a 1-year gap to avoid any trend to a particular day, precise campaign, user profile, or specific period (Sahu, 2021).

Attribute Information:

The dataset entails of ten (10) numerical and eight (8) categorical attributes.

Revenue: Class level. Possible values: False and True.

Administrative, Administrative Duration: Represent the administrative pages stop at by the visitor in that session and the total time spent in each of this page category.

Informational, Informational Duration: Represent the information related pages visited by the visitor in that session and total time spent in each of this page category.

Product Related and Product Related Duration: Represent the product related pages visited by the visitor in that session and total time spent in each of this page category.

Bounce Rate denotes to the percentage of visitors who enter the site from that page and then leave without activating any other requests to the analytics server during that session.

Exit Rate portrays the percentage of exits on a page.

Page Value part signifies the average value for a web page that a user visited before finalising an e-commerce transaction.

Special Day part shows the closeness of the site visiting time to a precise special day (Sahu, 2021).

The dataset also comprises of some other structures such as browser, operating system, region, traffic type, visitor type as returning or new visitor, a Boolean value representing whether the date of the visit is month and weekend of the year.